19.05.2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: Jan 2025– April 2025		
Maximum Marks: 100 Examination: ESE Examination		Duration: 3 Hrs.
Programme code: 01/40 Programme: B. Tech in COMP/ B. Tech in EXCP	Class: SY	Semester: III (SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: Computer Engineering /Electronics and Computer Engineering	
Course Code: 16U01C301 / 16U40C301	Name of the Course: Integral Transform and Vector Calculus	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four** of the following	**20**
i)	Find $L(te^{-t}\cosh 2t)$	05
ii)	Find Inverse Laplace Transform of $\frac{2s+3}{s^2+2s+2}$	05
iii)	Find Half range Sine series of $f(x) = 1 + x$ in the interval $(0,1)$	05
iv)	Find Z transform of $f(k)$ indicating the region of convergence where $f(k) = \left(\frac{1}{2}\right)^k, k \geq 0$	05
v)	Find the volume of the tetrahedron with vertices $(1,2,3), (4,5,6), (-1,3,4), (4,0,5)$	05
vi)	Find the total work done in moving a particle in the force field $\bar{F} = 3xyi - 5zj + 10xk$ along $x = t^2 + 1, y = 2t^2, z = t^3$ from $t = 1$ to $t = 2$.	05
Q2 A	Solve the following	**10**
i)	Find Laplace Transform of $\frac{\sin 2t}{t}$	05
ii)	Find Inverse Laplace Transform of $cot^{-1}\left(\frac{s}{2}\right)$	05
	OR	
Q2 A	Find the Fourier expansion of $f(x) = x + x^2$ when $-\pi \leq x \leq \pi$ and $f(x + 2\pi) = f(x)$. Hence deduce that $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\ldots\ldots$	10
Q 2 B	Solve any **One** of the following	**10**
i)	Using Laplace Transform Solve $(D^2 - 3D + 2)y = 4e^{2t}, y(0) = -3,\ y'(0) = 5.$	10

ii)	Find Fourier series for $f(x) = \frac{1}{2}(\pi - x)$ in $(0, 2\pi)$ Hence deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots\ldots$	10
Q3	Solve any **Two** of the following	**20**
i)	Find Complex form of Fourier series of $f(x) = \cosh 2x + \sinh 2x$ in the interval $(-2, 2)$	10
ii)	If $\Phi = x^3 + y^3 + z^3 - 3xyz$, find **(a)** $\bar{r} \cdot \nabla\Phi$ **(b)** div $\bar{F}$ **(c)** curl $\bar{F}$ where $\bar{F} = \nabla\Phi$.	10
iii)	Prove that $\bar{F} = (2xy + z)i + (x^2 + 2yz^3)j + (3y^2z^2 + x)k$ is irrotational. Find the scalar point function $\emptyset$ such that $\bar{F} = grad\ \emptyset$ and evaluate $\int_A^B \bar{F} \cdot d\bar{r}$ along the straight line joining $A(1,2,0)$ to $B(2,2,1)$.	10
Q4	Solve any **Two** of the following	**20**
i)	**(a)** Find the inverse Z- transform of $\frac{1}{(z-2)^2}$ for $\lvert z \rvert < 2$.	05
	(b) Use Gauss's divergence theorem to evaluate $\iint_s \hat{N} \cdot \bar{F}\, ds$ where $\bar{F} = 4x\, i + 3yj - 2zk$ and s is the closed surface bounded by $x = 0, y = 0, z = 0$ *and* $2x + 2y + z = 4$.	05
ii)	Find the directional derivative of $\Phi = x^4 + y^4 + z^4$ at point $A(1, -2, 1)$ in the direction of AB where B is $(2, 6, -1)$. In what direction the directional derivative of Φ is maximum? Also find the magnitude of the maximum directional derivative.	10
iii)	Verify Green's theorem for $\bar{F} = (xy + y^2)i + x^2 j$ and C is the closed curve of the region bounded by $y = x$ and $y = x^2$.	10
Q5	Solve any **Four** of the following	**20**
i)	Evaluate $\int_0^\infty e^{-2t}\,[(1 - 4t + 2t^2)H(t-1)]\, dt$	05
ii)	Find $L^{-1}\left(\frac{1}{(s-2)^2(s+3)}\right)$ using Convolution theorem.	05
iii)	Find Fourier Sine transform of $f(x) = e^{-2x}, x > 0$	05
iv)	Find the inverse Z- transform of $\frac{1}{(z-2)(z-3)}$ for $2 < \lvert z \rvert < 3$.	05
v)	Find a, b, c if $\bar{F} = (axy + bz^3)i + (3x^2 - cz)j + (3xz^2 - y)k$ is irrotational.	05
vi)	Use Stoke's theorem to evaluate $\int_C \bar{F} \cdot d\bar{r}$ where $\bar{F} = yz\hat{\imath} + zx\hat{\jmath} + xy\hat{k}$ and C is the boundary of the circle $x^2 + y^2 + z^2 = 1, z = 0$	05